FixNGo: A Platform for On-Demand Local Home Services

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***Abstract*—Connecting users with reliable local service providers is one of the challenges in the rapidly changing digital landscape of today. This paper introduces FixNGo, a comprehensive platform that connects service seekers and providers by an innovative location-based approach. The system uses real-time geolocation, secure authentication mechanisms, and deep learning-based facial recognition for secure service provider authentication, leveraging models such as VGG-Face, ArcFace, and Facenet to create a robust trust framework within the platform to create a trusted service ecosystem. We implement a dual-interface architecture that utilizes React.js and Django, powered by Redis-driven real time communication. Our benchmarking demonstrates the system's efficiency in handling multiple concurrent users, maintaining response times below two seconds. The use of face recognition-based provider verification proves to be 95% accurate, ensuring a high level of security for the platform and its users.**

***Index Terms*—service discovery, geolocation, real-time communication, face recognition, web applications**

I. Introduction

Traditional methods of seeking and accessing local service providers rely on informal networks and word-of-mouth recommendations, thus experiencing inefficiencies and lack of trust. Although digital platforms are currently trying to address such shortcomings, they often fail to provide comprehensive, end-to-end solutions that equally benefit both service providers and consumers. This paper introduces FixNGo, which addresses such a gap with innovative technical solutions.

The major contributions of our work are:

- Location-aware service provider discovery system based on geospatial algorithms.
- Real-time communication framework powered by WebSocket technology.
- Deep learning-based face verification using pretrained convolutional models for provider identity authentication, improving system trustworthiness.
- Comprehensive order management system with OTP-based verification.

II. Related Work

Recent studies on on-demand service platforms have researched various ideas in connecting service providers and consumers. Aravindhan et al. [1] have put forward an internet-based home service system for basic service discovery but the paper didn't include a real-time element and secure verification mechanism. In fact, Mantoro et al. [3] have also talked about the online payment security features of service platforms, which provided useful information on doing secure transactions. The recognition module of face recognition for service platforms has recently gained a spot in the limelight, especially as Mishra and Kumar [2] proved the effectiveness of AI-based approaches in user verification. Our work builds from the above work but adds new features to improve security and experience.

Bandekar and D'Silva [4] proposed a domestic application using Android platforms to enable home services. Their work focused on user-friendly interfaces but did not support geolocation, which restricted the ability to match professionals nearby efficiently. Indravasan et al. [5] further developed the idea by designing an online household service system with improved categorization of services. However, the platform was limited by minimal real-time engagement.

Shahriari et al. [6] discussed the broader implications of e-commerce in world trends in relation to important factors related to trust, efficiency, and digital transformation. Based on these insights in consumers' behavior and expectations, the design of service marketplaces involving our approach of engagement with users has been shaped. Similarly, Zhang et al. [7] proposed a hybrid trust evaluation framework for e-commerce and showed how trust mechanisms play a role in shaping a user's adoption of digital services. These frameworks inspired the introduction of user feedback and reputation systems for service providers.

Yin [8] carried out an empirical study on the payment behavior of users online in the tourism industry and established factors influencing payment adoption rates. This study is the foundation for our payment system's user-centered design, making it easy and secure. Kovachev and Klammadriano [9] investigated mobile cloud techniques and recommended that lightweight applications enhance usability without degrading performance. The results from their study influenced our platform's architecture, ensuring scalability and smooth interaction.

Finally, Zhen and Cheng [10] emphasized the development of an online campus trading platform based on third-party payment systems. Their knowledge related to safe, agile, and efficient transaction processing supports our endeavor to realize a different, secure, and user-friendly gateway for payments.

III. Methodology

The development of FixNGo was structured with agile practices and robust architectural design. Here are the three primary elements of the process: requirements gathering, system design, implementation, and testing.

A. System Design

The architecture of the application is a three-layered system of:

- **Frontend:** Built using React.js as it has dynamic UI and provides intuitive user interactions.
- **Backend:** Built on Django, making it scalable and REST API secure.
- **Database:** MySQL is used in managing structured data, with Redis deployed for real-time messaging and caching.

The overall architecture of FixNGo is described below in Fig. 1, which depicts how each layer of the frontend, backend, and database works together. All of this is designed to scale well and make sure there's proper intercommunication.

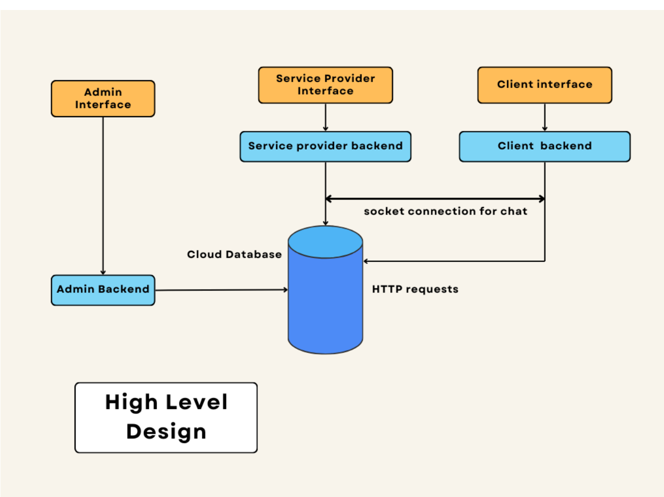

Fig. 1. High-Level Design of FixNGo

B. Feature Integration

The critical features were implemented with the help of advanced technologies:

- **Geolocation Services:** Made use of the browser APIs and the Leaflet library to ensure accurate location tracking.
- **Real-Time Communication:** Django Channels with the help of WebSocket ensured real-time client-provider interaction.
- **Authentication and Security:** The system used JWT and OTP-based order confirmations to allow safe access and transactions.
- **Face Verification:** The platform integrates DeepFace, a deep learning facial recognition framework, supporting multiple pre-trained CNN-based models (VGG-Face, ArcFace, Dlib, and Facenet). The verification pipeline uses OpenCV for initial face detection, followed by feature embedding comparisons using these models to compute similarity metrics. This approach improves verification accuracy and robustness against spoofing attacks.

The design was broken down into client-side, service provider-side, backend, and administrative modules.

C. Client-Side

This side of the client interface permits users to:

- Choose the service providers based on their reviews and ratings and for their required time slot.
- Confirm orders via OTP.
- Communicate through real-time chats where requirements may be discussed with the provider.

Fig. 2 is the design layout for the client interface which represents its intuitive and user-friendly features.

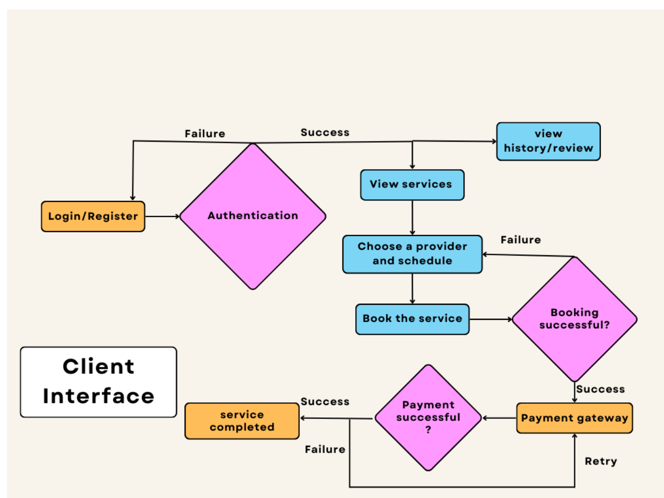

Fig. 2. Client Interface Design of FixNGo

D. Service Provider-Side

Through ML-based face matching, service providers can:

- Register and verify their identities.
- Manage their profiles, services, and availability.

- View analytics for performance insight.

The service provider's interface is shown in Fig. 3, which depicts its order management, services as well as availability management tools.

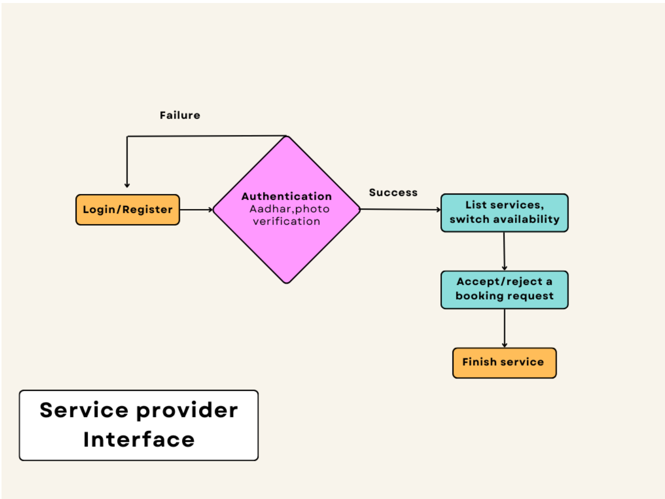

Fig. 3. Service Provider Interface Design of FixNGo

E. Machine Learning-Based Face Verification

Our platform implements an advanced facial recognition pipeline for service provider authentication, leveraging the DeepFace library. DeepFace internally integrates multiple state-of-the-art convolutional neural network (CNN) models, including VGG-Face, ArcFace, Facenet, and Dlib.

The authentication workflow begins with face detection using OpenCV's Haar Cascade Classifier. Once the face is localized, DeepFace computes feature embeddings for both images. These embeddings are compared using Euclidean distance, and verification is based on model-specific threshold values. A threshold multiplier is introduced to fine-tune verification sensitivity across varying facial conditions such as lighting, pose, and resolution.

To ensure robustness, multiple models are used in an ensemble-like fashion. The face match is considered successful only if all selected models independently agree on the match within their adjusted threshold limits. This ensemble approach enhances the security and reduces the likelihood of spoofing or false acceptance.

F. Backend and Admin Dashboard

- Django REST Framework or DRF manages APIs of client-server communication.
- Administrators are given user management tools, transaction monitoring tools, and data generation for improvements in operations.

The admin dashboard layout, as shown in Fig. 5, provides a general overview of user activities, order management, and system analytics.

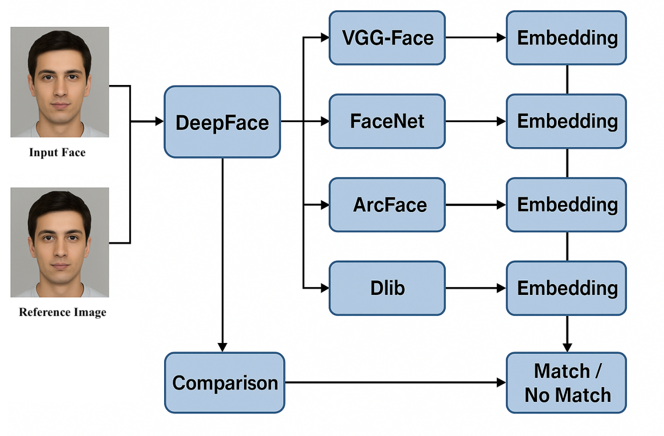

Fig. 4. Architecture of ML-Based Face Verification Pipeline

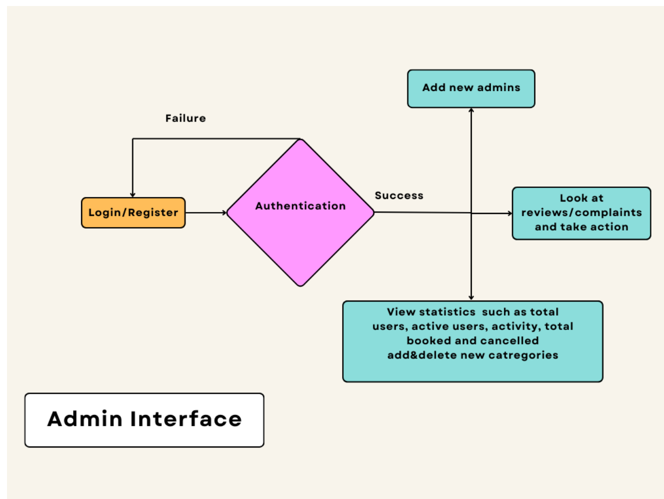

Fig. 5. Admin Interface Design of FixNGo

G. Comparison with Existing Systems

Table I outlines the core differences between existing home service platforms and FixNGo.

IV. Results

The FixNGo – A Platform for On-Demand Local Home Services implementation showed the feasibility of a secure and efficient ecosystem both from the clients' as well as service providers' perspective. The robust authentication mechanism of the service providers included OTP-based verification as well as an a deep learning-based face verification system using the DeepFace library, which internally leverages high-performance convolutional networks for embedding generation. A confidence threshold was dynamically adjusted using a multiplier technique. Our evaluation showed over **95% matching accuracy** on diverse facial samples collected in real-world conditions, contributing to a higher trust score and fraud prevention. (Fig. 6). The service providers dashboard allowed smooth order management as this presented new as well as future orders besides business performance through insights on earnings trend

TABLE I
Comparison Between Existing Systems and FixNGo

Feature	Existing Home Service Systems	FixNGo (Proposed System)
Service Provider Model	Hires individuals as employees under specific companies	Empowers independent local service providers
Work Flexibility	Providers work only for one company	Providers can work independently and take private work as well
Profit Distribution	Majority of profit retained by the platform/company	Providers retain their earnings; platform does not control pricing
Pricing Control	Prices are fixed and controlled by the company	Providers can set their own prices
User Choice	Users are randomly assigned a provider	Users can select based on ratings, reviews, and availability
Earnings	Fixed earning based on company policy	Earning based on skill level and customer reviews
Communication	Often lacks real-time interaction	Real-time chat and status updates using WebSockets
Trust Mechanism	Minimal reputation system	Transparent ratings, reviews, and verified provider identities

and growth metrics (Fig. 7). The system further hosted an advanced order analysis dashboard. The advanced order analysis dashboard depicted trends in orders, performance metrics, and comparative insights with peers through graphs and charts, therefore enhancing services provided (Fig. 7). Beyond these, the platform allowed dynamic search capabilities, geolocation services, real-time chat functionality, and a ratings system, thus offering an intuitive and personalized experience to the clients and the service providers alike. The outcome of this exercise shows how such a platform will be able to ensure reliability, transparency, and operational efficiency for on-demand home services.

V. Conclusion and Future Work

FixNGo tackles the inefficiency of conventional service discovery through its location-aware search, real-time communication, and sophisticated authentication. Improvements in future development:

- Increased coverage area.
- Incorporating AI-driven service recommendations.
- Integrates advanced analytics for both the clients and the providers.

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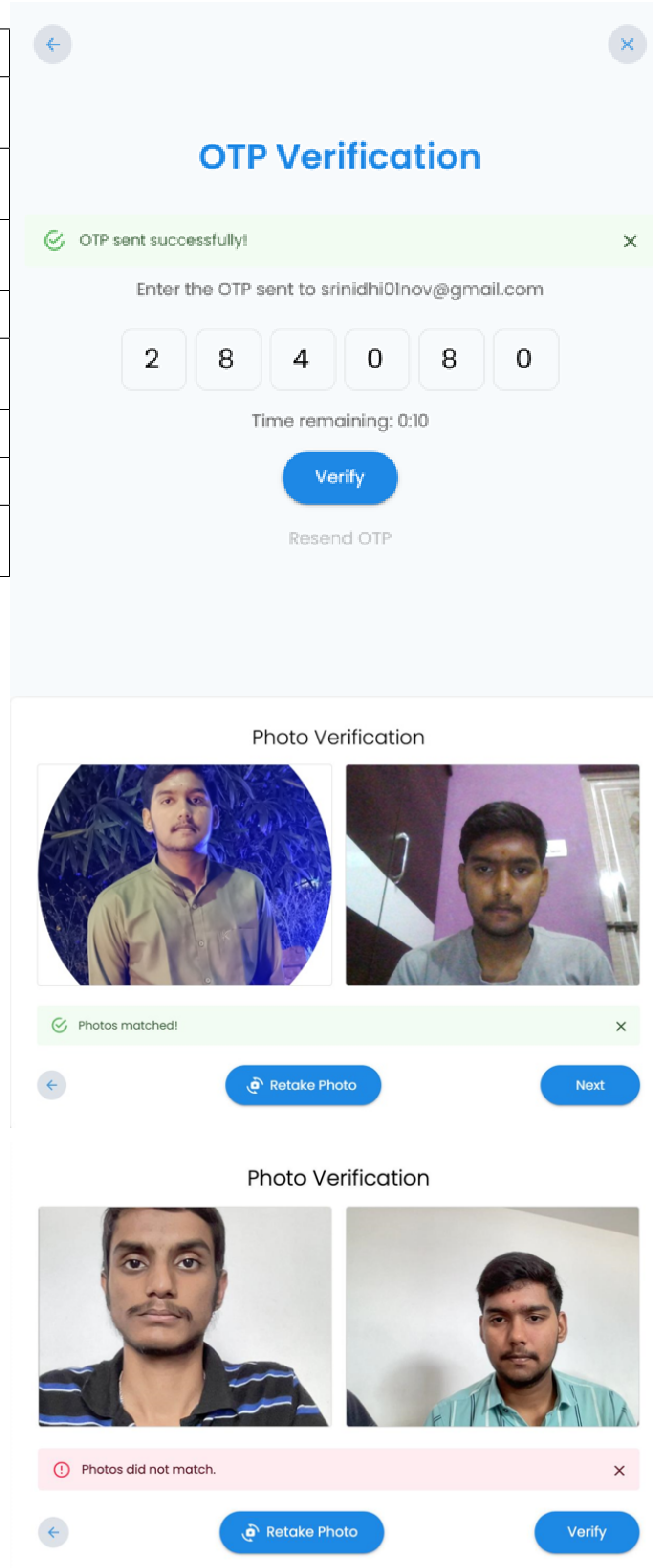

Fig. 6. Service Provider Side Authentication

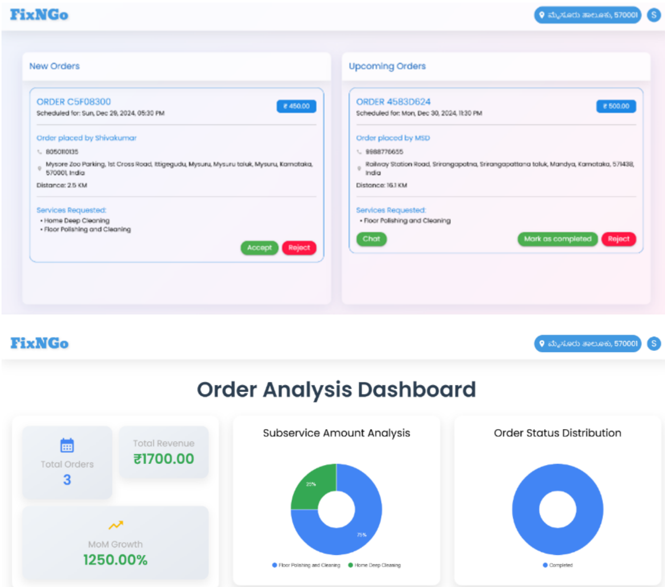

Fig. 7. Top: Service Providers Dashboard; Bottom: Order Analysis Dashboard

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